17. Windows OS Commands Execution (ping, ifconfig, netstat)

Aim

To execute and understand Windows OS networking commands such as `tracert`, `ping`, `ipconfig` (Windows alternative to `ifconfig`), and `netstat` to check network connectivity, IP configuration, and active network connections.

Procedure

1. Start the Windows Command Prompt by pressing `Win + R`, typing `cmd`, and pressing Enter.
2. Execute the `ping` command to check connectivity to a host.
3. Execute the `tracert` command to trace the route to a destination.
4. Execute the `ipconfig` command to display IP configuration.
5. Execute the `netstat` command to display active network connections.
6. Observe the output of each command.

Code / Commands

```
ping google.com
ipconfig
netstat -an
```

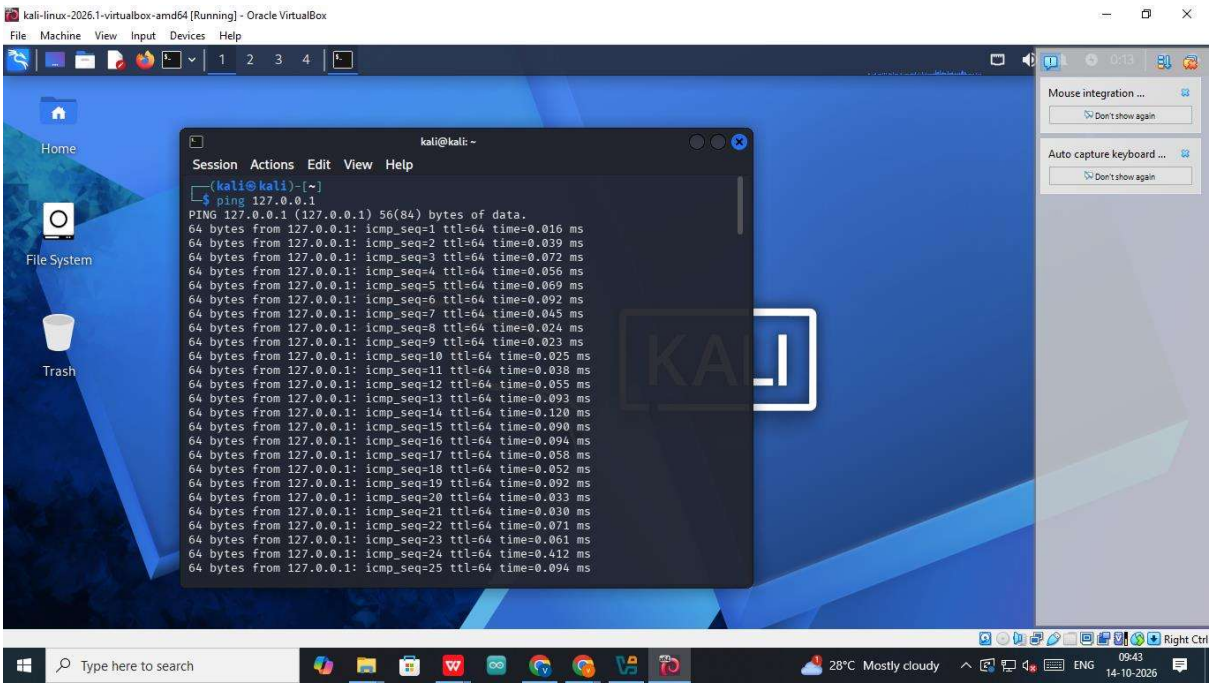
Sample Input and Output

Command 1: ping

Sample Input:

```
ping 127.0.0.1
```

Sample Output:

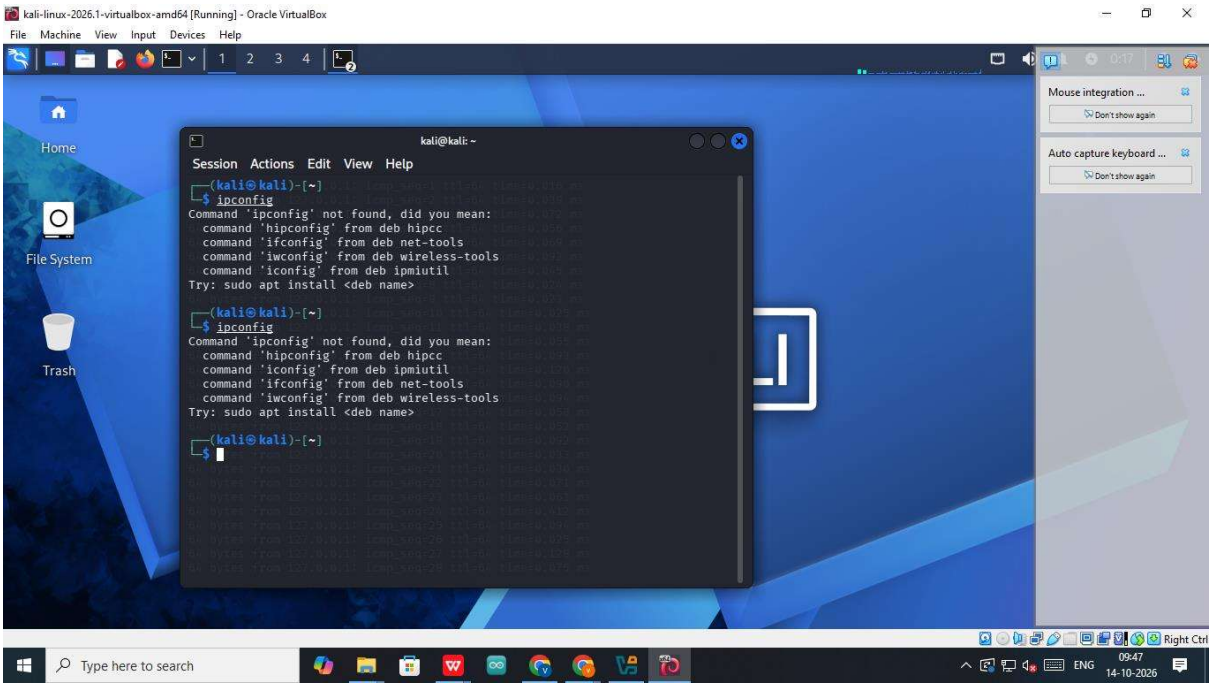


Command 2: ipconfig

Sample Input:

```
ipconfig
```

Sample Output:

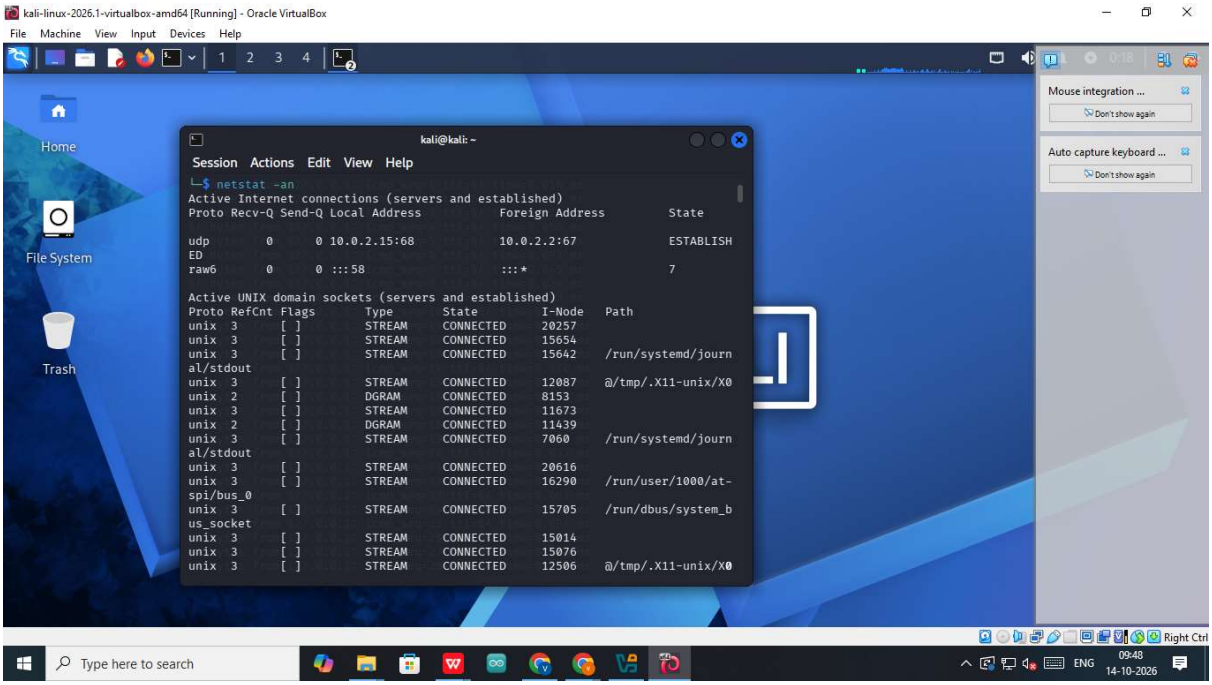


Command 3: netstat

Sample Input:

```
netstat -an
```

Sample Output:



Result

Thus, the Windows networking commands `ping`, `ipconfig`, and `netstat` were studied and executed to check network connectivity, trace routes, view IP configuration, and identify active network connections.